

Division Decoded

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Two Kinds of Division

Facts first, then split the dividend into friendly pieces.

Warm-Up 12 ITEMS • RECALL

1 $42 \div 6$

2 $56 \div 8$

3 $63 \div 9$

4 $36 \div 4$

5 $72 \div 8$

6 $45 \div 5$

7 $12 \div 3$

8 $20 \div 4$

9 $18 \div 2$

10 $25 \div 5$

11 $30 \div 10$

12 $16 \div 4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $84 \div 4$ ($80 \div 4 = 20$, then $4 \div 4 = 1$)
.....

14. $96 \div 6$ ($60 \div 6 = 10$, then $36 \div 6 = 6$)
.....

15. $75 \div 5$ ($50 \div 5 = 10$, then $25 \div 5 = 5$)
.....

16. $90 \div 6$ ($60 \div 6 = 10$, then $30 \div 6 = 5$)
.....

17. $78 \div 3$ ($60 \div 3 = 20$, then $18 \div 3 = 6$)
.....

18. $56 \div 7$
.....

19. $48 \div 6$
.....

20. $81 \div 9$
.....

The Missing Side

Area and one side are given. Break the area up and hunt the other side.

✦ Warm-Up 12 ITEMS • RECALL

1 $24 \div 3$

2 $54 \div 6$

3 $81 \div 9$

4 $32 \div 8$

5 $49 \div 7$

6 $60 \div 5$

7 $14 \div 2$

8 $21 \div 3$

9 $28 \div 4$

10 $35 \div 5$

11 $40 \div 8$

12 $27 \div 9$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $128 \div 4$ ($120 \div 4 = 30$, then $8 \div 4 = 2$)
.....

14. $156 \div 6$ ($120 \div 6 = 20$, then $36 \div 6 = 6$)
.....

15. $144 \div 3$ ($120 \div 3 = 40$, then $24 \div 3 = 8$)
.....

16. $175 \div 5$ ($150 \div 5 = 30$, then $25 \div 5 = 5$)
.....

17. $168 \div 7$ ($140 \div 7 = 20$, then $28 \div 7 = 4$)
.....

18. $84 \div 7$
.....

19. $96 \div 8$
.....

20. $108 \div 9$
.....

21. $66 \div 6$
.....

22. $77 \div 7$
.....



Working space



Error journal

When It Doesn't Fit

Quotient and remainder are two different answers. Read which one is being asked for.

Warm-Up 6 ITEMS • RECALL

1 $10 \div 5$

2 $24 \div 6$

3 $36 \div 6$

4 $45 \div 5$

5 $32 \div 8$

6 $50 \div 10$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

7. $17 \div 5 \rightarrow$ quotient (3 fives is 15, and 2 are left) _____

8. $94 \div 4 \rightarrow$ quotient ($80 \div 4 = 20$, then $12 \div 4 = 3$) _____

9. $137 \div 6 \rightarrow$ quotient ($120 \div 6 = 20$, then $12 \div 6 = 2$) _____

10. $92 \div 4$ _____

11. $78 \div 6$ _____

12. $85 \div 5$ _____

13. $126 \div 9$ _____

14. $115 \div 5$ _____

★

Challenge

NOT GRADED · REASON IT OUT

15. $17 \div 5 \rightarrow$ remainder
-
16. $23 \div 4 \rightarrow$ remainder
-
17. $38 \div 6 \rightarrow$ remainder
-
18. $50 \div 7 \rightarrow$ remainder
-
19. $29 \div 3 \rightarrow$ remainder
-
20. $45 \div 8 \rightarrow$ remainder
-
21. $94 \div 4 \rightarrow$ remainder ($4 \times 23 = 92$)
-
22. $137 \div 6 \rightarrow$ remainder ($6 \times 22 = 132$)
-
23. 26 kids, vans hold 6. Vans needed? (Four vans carry 24 – two kids still need a seat)
-
24. 26 cookies for 6 friends. Whole cookies each? (This time the leftover stays on the plate)
-
25. $288 \div 12$
-
26. $384 \div 16$
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Chunks Get Tidy

Bigger dividends. Pull out the biggest easy chunk first, then finish.

✦ Warm-Up 12 ITEMS • RECALL

1 $120 \div 4$

2 $150 \div 5$

3 $240 \div 6$

4 $350 \div 7$

5 $180 \div 9$

6 $160 \div 8$

7 $9 \div 3$

8 $15 \div 5$

9 $22 \div 2$

10 $33 \div 3$

11 $44 \div 4$

12 $55 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $372 \div 3$ ($300 \div 3 = 100$, $60 \div 3 = 20$, $12 \div 3 = 4$)
.....

14. $465 \div 5$ ($450 \div 5 = 90$, then $15 \div 5 = 3$)
.....

15. $728 \div 8$ ($720 \div 8 = 90$, then $8 \div 8 = 1$)
.....

16. $594 \div 6$ ($540 \div 6 = 90$, then $54 \div 6 = 9$)
.....

17. $952 \div 7$ ($700 \div 7 = 100$, $210 \div 7 = 30$, $42 \div 7 = 6$)
.....

18. $144 \div 6$
.....

19. $168 \div 8$
.....

20. $175 \div 7$
.....

21. $192 \div 6$
.....

22. $225 \div 9$
.....



Working space



Error journal

Remainder Race & Puzzles

Enrichment day. Every leftover needs a decision — say which one you made.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦

Warm-Up

2 ITEMS • RECALL

1 18 ÷ 6

2 42 ÷ 7

◆

Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 91 ÷ 7

.....

4. 120 ÷ 8

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 1 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Two Kinds of Division

OUT OF 20

Warm-Up • 1-12 • 7 | 7 | 7 | 9 | 9 | 9 | 4 | 5 | 9 | 5 | 3 | 4

Core • 13-20 • 21 | 16 | 15 | 15 | 26 | 8 | 8 | 9

Challenge • 21-24 • 16 | 9 | 12 | 12

1.2 • The Missing Side

OUT OF 22

Warm-Up • 1-12 • 8 | 9 | 9 | 4 | 7 | 12 | 7 | 7 | 7 | 7 | 5 | 3

Core • 13-22 • 32 | 26 | 48 | 35 | 24 | 12 | 12 | 12 | 11 | 11

Challenge • 23-26 • 36 | 24 | 13 | 14

1.3 • When It Doesn't Fit

OUT OF 14

Warm-Up • 1-6 • 2 | 4 | 6 | 9 | 4 | 5

Core • 7-14 • 3 | 23 | 22 | 23 | 13 | 17 | 14 | 23

Challenge • 15-26 • 2 | 3 | 2 | 1 | 2 | 5 | 2 | 5 | 5 | 4 | 24 | 24

1.4 • Chunks Get Tidy

OUT OF 22

Warm-Up • 1-12 • 30 | 30 | 40 | 50 | 20 | 20 | 3 | 3 | 11 | 11 | 11 | 11

Core • 13-22 • 124 | 93 | 91 | 99 | 136 | 24 | 21 | 25 | 32 | 25

Challenge • 23-26 • 211 | 1 | 63 | 72

Fri • Remainder Race & Puzzles

OUT OF 4

Warm-Up • 1-2 • 3 | 6

Core • 3-4 • 13 | 15

Challenge • 5-15 • 14 | 2 | 13 | 84 | 7 | 6 | 56 | 12 | 36 | 25 | 125

Round It Up

Vans, boats and buses. Two leftovers still force one more of everything.

Warm-Up 7 ITEMS • RECALL

1 $20 \div 5$

2 9 people, vans of 4 — vans needed

3 12 people, vans of 4

4 13 people, vans of 4

5 7 people, boats of 3

6 $11 \div 3$ — the whole part

7 $26 \div 5$ — the whole part

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. 127 children, buses of 30 — buses needed

.....

9. 98 books, boxes of 12 — boxes needed

.....

10. 53 eggs, cartons of 6 — cartons needed

.....

11. 200 seats, rows of 18 — rows needed

.....

12. 75 guests, tables of 8 — tables needed

.....

13. $83 \div 9$ — the whole part

.....

14. $94 \div 7$ — the whole part

.....

★

Challenge

NOT GRADED · REASON IT OUT

15. $21 \div 5$ — the remainder
-
16. A bus holds 44. How many for 500 children
-
17. If $127 \div 30$ needs 5 buses, how many seats sit empty
-
18. $13 \div 2$ — the remainder
-
19. $17 \div 3$ — the remainder
-
20. $22 \div 4$ — the remainder
-
21. $19 \div 5$ — the remainder
-
22. $47 \div 6$ — the remainder
-
23. $59 \div 7$ — the remainder
-
24. $75 \div 8$ — the remainder
-
25. $250 \div 12$ — the remainder
-
26. $365 \div 7$ — the remainder
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Drop It

Whole ones each. The leftover just stays on the plate.

Warm-Up 9 ITEMS • RECALL

1 $26 \div 6$ — whole cookies each

2 $26 \div 6$ — left on the plate

3 $17 \div 5$ — whole each

4 $30 \div 7$ — whole each

5 $30 \div 7$ — leftover

6 $19 \div 4$ — whole each

7 $20 \div 3$ — the whole part

8 $14 \div 4$ — the whole part

9 $25 \div 6$ — the whole part

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. \$50 shared by 8 — whole dollars each

.....

11. \$50 shared by 8 — dollars left

.....

12. $145 \div 12$ — whole each

.....

13. $145 \div 12$ — leftover

.....

14. $200 \div 30$ — whole each

.....

15. $100 \div 7$ — the whole part

.....

16. $137 \div 9$ — the whole part

.....

★ Challenge NOT GRADED • REASON IT OUT

17. 90 cm of ribbon into 14 cm bows — whole bows

18. That ribbon — cm wasted

19. $20 \div 3$ — the remainder

20. $14 \div 4$ — the remainder

21. $9 \div 2$ — the remainder

22. $100 \div 7$ — the remainder

23. $137 \div 9$ — the remainder

24. $88 \div 6$ — the remainder

25. $500 \div 24$ — the whole part

26. $500 \div 24$ — the remainder

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Share It Out

Cut the leftovers up and the remainder becomes a fraction.

Warm-Up 7 ITEMS • RECALL

1 $9 \div 2$ — whole part

2 7 kids share 15 apples — each gets

3 ... and left over

4 4 kids share 10 — each

5 ... left over

6 3 kids share 12 — each

7 5 kids share 22 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. $17 \div 4$ — type the fraction part as a/b

.....

9. \$45 shared by 4 — dollars each

.....

10. 6 vans hold 8, 50 kids — vans needed

.....

11. Boxes of 12, 100 items — full boxes

.....

12. ... items left over

.....

13. 9 seats a row, 60 people — rows needed

.....

14. Bags of 6, 45 marbles — full bags

.....

The Remainder Is the Answer

Clocks and repeating patterns. Sometimes the leftover is the whole point.

Warm-Up 7 ITEMS • RECALL

1 Days in a week

2 $8 \div 2$

3 $12 \div 4$

4 $15 \div 3$

5 $18 \div 9$

6 $24 \div 8$

7 $30 \div 6$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

8. Pattern red, blue, green repeating. The 10th bead — type 1, 2 or 3 for its place in the group

9. It is 9 o'clock. 25 hours later, the hour shown

10. $105 \div 5$

11. $128 \div 4$

12. $147 \div 7$

13. $162 \div 6$

14. $189 \div 9$

Snack Run Planning

Real quantities, real people. Every leftover decision written down.

✦ Warm-Up 4 ITEMS • RECALL

1 24 crackers between 4 — each

2 20 grapes between 3 — whole each

3 $16 \div 8$

4 $21 \div 7$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A pack of 18 between 5 — whole each
.....

6. A pack of 18 between 5 — leftover
.....

7. $98 \div 7$
.....

8. $117 \div 9$
.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Round It Up

OUT OF 14

Warm-Up • 1-7 • 4 | 3 | 3 | 4 | 3 | 3 | 5

Core • 8-14 • 5 | 9 | 9 | 12 | 10 | 9 | 13

Challenge • 15-26 • 1 | 12 | 23 | 1 | 2 | 2 | 4 | 5 | 3 | 3 | 10 | 1

2.2 • Drop It

OUT OF 16

Warm-Up • 1-9 • 4 | 2 | 3 | 4 | 2 | 4 | 6 | 3 | 4

Core • 10-16 • 6 | 2 | 12 | 1 | 6 | 14 | 15

Challenge • 17-26 • 6 | 6 | 2 | 2 | 1 | 2 | 2 | 4 | 20 | 20

2.3 • Share It Out

OUT OF 14

Warm-Up • 1-7 • 4 | 2 | 1 | 2 | 2 | 4 | 4

Core • 8-14 • $\frac{1}{4}$ | 11.25 | 7 | 8 | 4 | 7 | 7

Challenge • 15-26 • 0.5 | 0.25 | 4.5 | 3.5 | 0.75 | 6.5 | 3.25 | 4.4 | $\frac{3}{4}$ | 12.5 | 7 | 13

2.4 • The Remainder Is the Answer

OUT OF 14

Warm-Up • 1-7 • 7 | 4 | 3 | 5 | 2 | 3 | 5

Core • 8-14 • 1 | 10 | 21 | 32 | 21 | 27 | 21

Challenge • 15-26 • 1 | 0 | 1 | 2 | 2 | 2 | 2 | 1 | 3 | 6 | 57 | 81

Fri • Snack Run Planning

OUT OF 8

Warm-Up • 1-4 • 6 | 6 | 2 | 3

Core • 5-8 • 3 | 3 | 14 | 13

Challenge • 9-14 • 2.4 | 5 | 8 | 64 | 69 | 143

Under the Bar

Partial quotients folded into the standard layout, line by line.

✦ Warm-Up 9 ITEMS • RECALL

1 $60 \div 3$

2 $80 \div 4$

3 $90 \div 3$

4 $48 \div 4$

5 $66 \div 6$

6 $84 \div 4$

7 $40 \div 4$

8 $120 \div 4$

9 $150 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

10. $96 \div 4$

11. $126 \div 3$

12. $175 \div 5$

13. $192 \div 6$

14. $133 \div 7$

15. $168 \div 4$

16. $216 \div 6$

17. $294 \div 7$

18. $312 \div 8$

19. $378 \div 9$

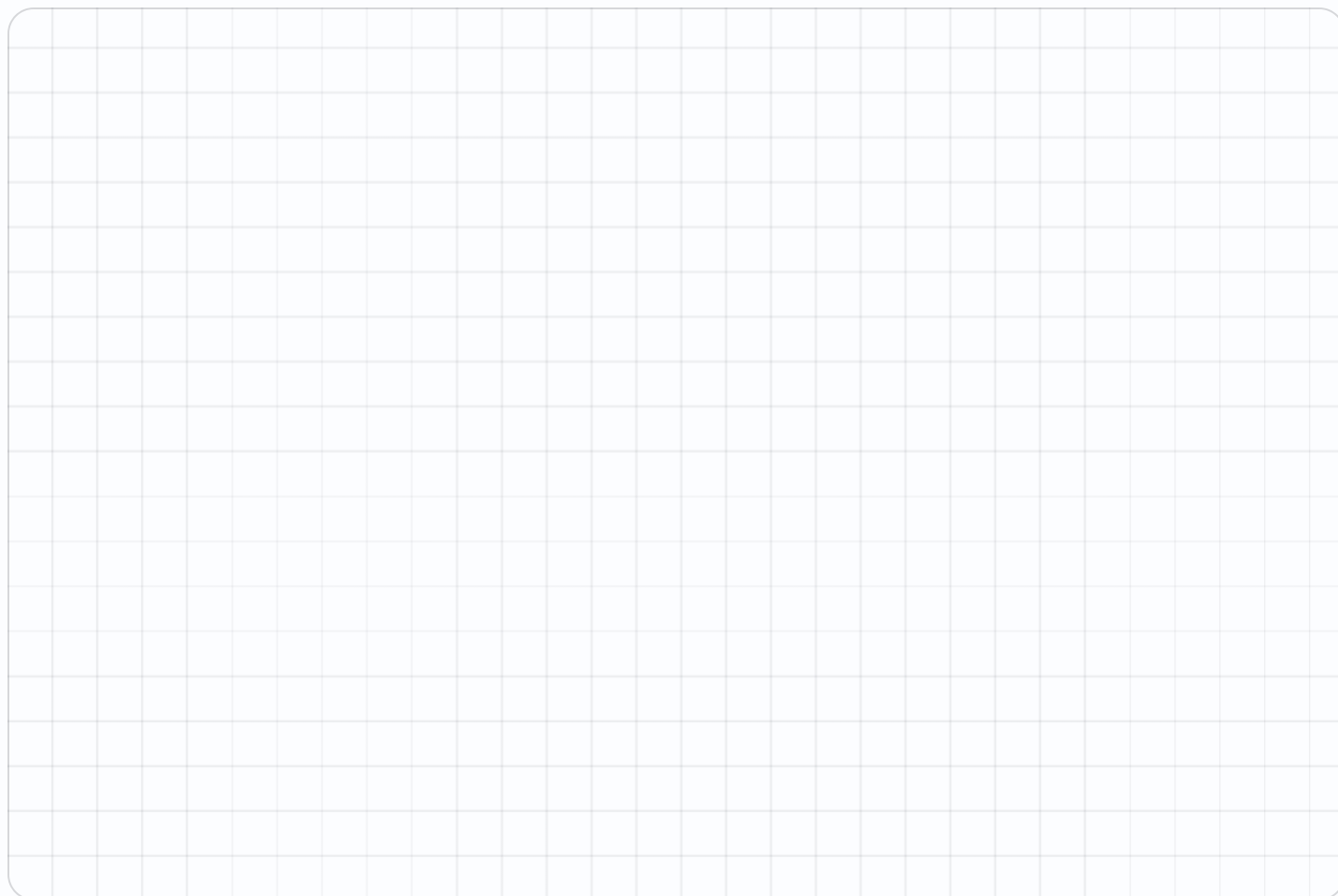
★ Challenge NOT GRADED • REASON IT OUT

20. $288 \div 8$
.....

21. $441 \div 9$
.....

22. $864 \div 6$
.....

23. $972 \div 4$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Three-Digit Dividends

Estimate before you start. Every time, out loud.

✦ Warm-Up 12 ITEMS • RECALL

1 $300 \div 3$

2 $400 \div 4$

3 $360 \div 6$

4 $250 \div 5$

5 $280 \div 7$

6 $540 \div 9$

7 $100 \div 10$

8 $200 \div 10$

9 $300 \div 100$

10 $500 \div 5$

11 $600 \div 6$

12 $800 \div 4$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $372 \div 3$

14. $456 \div 4$

15. $595 \div 5$

16. $728 \div 8$

17. $651 \div 7$

18. $420 \div 6$

19. $560 \div 8$

20. $630 \div 9$

21. $720 \div 8$

22. $810 \div 9$



Zeros in the Quotient

The place nobody remembers to hold. It catches everybody once.

✦ Warm-Up 12 ITEMS • RECALL

1 $60 \div 6$

2 $90 \div 9$

3 $200 \div 2$

4 $606 \div 6$

5 $404 \div 4$

6 $800 \div 8$

7 $22 \div 2$

8 $36 \div 3$

9 $52 \div 4$

10 $70 \div 5$

11 $90 \div 6$

12 $112 \div 7$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $618 \div 6$

14. $721 \div 7$

15. $915 \div 3$

16. $832 \div 8$

17. $525 \div 5$

18. $243 \div 3$

19. $364 \div 4$

20. $455 \div 5$

21. $546 \div 6$

22. $637 \div 7$

Check by Multiplying

Quotient $\times$ divisor + remainder = dividend. Self-marking from here on.

✦ Warm-Up 10 ITEMS • RECALL

1 7×8

2 $56 \div 8$

3 $9 \times 6 + 2$

4 $48 \div 4$

5 $6 \div 6$

6 $0 \div 5$

7 $7 \div 1$

8 $11 \div 11$

9 $9 \div 1$

10 $0 \div 12$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $106 \div 6$ — the quotient
.....

12. $330 \div 3$
.....

13. $440 \div 4$
.....

14. $550 \div 5$
.....

15. $606 \div 6$
.....

16. $707 \div 7$
.....

 Challenge NOT GRADED • REASON IT OUT

17. 12×4

18. $11 \times 5 + 3$

19. 124×3 — the check for $372 \div 3$

20. $17 \times 6 + 4$

21. $106 \div 6$ — the remainder

22. 91×8

23. A quotient of 23 remainder 5 with divisor 9 — the dividend

24. A quotient of 47 remainder 3 with divisor 12 — the dividend

25. $5050 \div 50$

26. $8080 \div 8$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

1 42 ÷ 6

2 90 ÷ 3

3 27 ÷ 3

4 48 ÷ 8

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 372 ÷ 3

6. 618 ÷ 6

7. 127 children, buses of 30 — buses needed

8. 252 ÷ 6

9. 336 ÷ 7

Challenge NOT GRADED • REASON IT OUT

10. $26 \div 4$ as a decimal

11. $100 \div 7$ — the remainder

12. A quotient of 23 remainder 5 with divisor 9 — the dividend

13. $1440 \div 12$

14. $2500 \div 25$

15. $999 \div 3$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Under the Bar

OUT OF 19

Warm-Up • 1-9 • 20 | 20 | 30 | 12 | 11 | 21 | 10 | 30 | 30

Core • 10-19 • 24 | 42 | 35 | 32 | 19 | 42 | 36 | 42 | 39 | 42

Challenge • 20-23 • 36 | 49 | 144 | 243

3.2 • Three-Digit Dividends

OUT OF 22

Warm-Up • 1-12 • 100 | 100 | 60 | 50 | 40 | 60 | 10 | 20 | 3 | 100 | 100 | 200

Core • 13-22 • 124 | 114 | 119 | 91 | 93 | 70 | 70 | 70 | 90 | 90

Challenge • 23-26 • 117 | 144 | 300 | 600

3.3 • Zeros in the Quotient

OUT OF 22

Warm-Up • 1-12 • 10 | 10 | 100 | 101 | 101 | 100 | 11 | 12 | 13 | 14 | 15 | 16

Core • 13-22 • 103 | 103 | 305 | 104 | 105 | 81 | 91 | 91 | 91 | 91

Challenge • 23-26 • 510 | 1006 | 91 | 91

3.4 • Check by Multiplying

OUT OF 16

Warm-Up • 1-10 • 56 | 7 | 56 | 12 | 1 | 0 | 7 | 1 | 9 | 0

Core • 11-16 • 17 | 110 | 110 | 110 | 101 | 101

Challenge • 17-26 • 48 | 58 | 372 | 106 | 4 | 728 | 212 | 567 | 101 | 1010

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 7 | 30 | 9 | 6

Core • 5-11 • 124 | 103 | 5 | 6.5 | 2 | 42 | 48

Challenge • 12-15 • 212 | 120 | 100 | 333

Tests for 2, 5 and 10

The last digit is enough on its own. Say why.

Warm-Up 12 ITEMS • RECALL

1 Is 46 divisible by 2 — yes or no

2 Is 45 divisible by 5

3 Is 45 divisible by 2

4 Is 70 divisible by 10

5 Is 75 divisible by 10

6 Is 128 divisible by 2

7 $14 \div 7$

8 $24 \div 3$

9 $32 \div 4$

10 $54 \div 6$

11 $64 \div 8$

12 $45 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

13. Is 3,450 divisible by 10 — yes or no
.....

14. Is 1,275 divisible by 5
.....

15. Is 2,346 divisible by 2
.....

16. Numbers below 50 divisible by both 2 and 5
.....

17. Is 999 divisible by 5
.....

18. $238 \div 7$
.....

19. $272 \div 8$
.....

20. $306 \div 9$
.....

21. $204 \div 6$
.....

22. $170 \div 5$
.....



Working space



Error journal

Tests for 3 and 9

Add the digits — then find out why that ridiculous trick works.

Warm-Up 12 ITEMS • RECALL

1 Digit sum of 24

2 Is 24 divisible by 3 — yes or no

3 Digit sum of 45

4 Is 45 divisible by 9

5 Digit sum of 17

6 Is 17 divisible by 3

7 Half of 20

8 Half of 36

9 A quarter of 20

10 A third of 21

11 A fifth of 25

12 Half of 50

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

13. Digit sum of 738

.....

14. Is 738 divisible by 9 — yes or no

.....

15. Is 1,234 divisible by 3

.....

16. Is 5,346 divisible by 9

.....

17. Digit sum of 9,999

.....

18. A quarter of 96

.....

19. A third of 111

.....

20. A fifth of 145

.....

21. An eighth of 96

.....

22. A sixth of 144

.....



Working space



Error journal

Four-Digit Dividends

Same method, longer bar. Do not flinch.

Warm-Up 12 ITEMS • RECALL

1 $1,000 \div 2$

2 $2,000 \div 4$

3 $3,000 \div 3$

4 $4,800 \div 8$

5 $5,600 \div 7$

6 $1,200 \div 6$

7 \$20 shared by 4

8 \$30 shared by 5

9 \$18 shared by 3

10 \$40 shared by 8

11 \$60 shared by 6

12 \$100 shared by 10

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3,472 \div 8$

14. $4,536 \div 6$

15. $5,432 \div 4$

16. $6,741 \div 3$

17. $8,127 \div 9$

18. \$144 shared by 12

19. \$175 shared by 7

20. \$210 shared by 6

21. \$252 shared by 9

22. \$180 shared by 15

★

Challenge

NOT GRADED · REASON IT OUT

23. 9,072 ÷ 8

24. 7,595 ÷ 5

25. \$1000 shared by 8

26. \$1440 shared by 24

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Two-Digit Divisors

Grade-five ceiling. Estimate, adjust, estimate again.

Warm-Up 12 ITEMS • RECALL

1 $40 \div 20$

2 $60 \div 30$

3 $90 \div 30$

4 $80 \div 20$

5 $120 \div 40$

6 $150 \div 50$

7 $12 \div 6$

8 $20 \div 5$

9 $28 \div 7$

10 $40 \div 10$

11 $36 \div 4$

12 $49 \div 7$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $84 \div 12$

14. $96 \div 16$

15. $224 \div 14$

16. $375 \div 25$

17. $391 \div 17$

18. $456 \div 6$

19. $532 \div 7$

20. $608 \div 8$

21. $684 \div 9$

22. $380 \div 5$

Perfect Numbers

$6 = 1 + 2 + 3$. Its factors add up to itself. Hunt the next one.

✦ Warm-Up 4 ITEMS • RECALL

1 Factors of 6 below 6 —
their sum

2 Factors of 8 below 8 —
their sum

3 $33 \div 11$

4 $60 \div 12$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Factors of 28 below 28 — their sum

.....

6. Factors of 12 below 12 — their sum

.....

7. $441 \div 7$

.....

8. $512 \div 4$

.....

 Challenge NOT GRADED • REASON IT OUT

9. The second perfect number

10. The third perfect number

11. Factors of 496 below 496 — their sum

12. $2016 \div 12$

13. $1728 \div 12$

14. $3025 \div 55$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Tests for 2, 5 and 10

OUT OF 22

Warm-Up • 1-12 • yes | yes | no | yes | no | yes | 2 | 8 | 8 | 9 | 8 | 5

Core • 13-22 • yes | yes | yes | 4 | no | 34 | 34 | 34 | 34 | 34

Challenge • 23-26 • 10 | 19 | 34 | 34

4.2 • Tests for 3 and 9

OUT OF 22

Warm-Up • 1-12 • 6 | yes | 9 | yes | 8 | no | 10 | 18 | 5 | 7 | 5 | 25

Core • 13-22 • 18 | yes | no | yes | 36 | 24 | 37 | 29 | 12 | 24

Challenge • 23-26 • 0 | 3 | 24 | 111

4.3 • Four-Digit Dividends

OUT OF 22

Warm-Up • 1-12 • 500 | 500 | 1000 | 600 | 800 | 200 | 5 | 6 | 6 | 5 | 10 | 10

Core • 13-22 • 434 | 756 | 1358 | 2247 | 903 | 12 | 25 | 35 | 28 | 12

Challenge • 23-26 • 1134 | 1519 | 125 | 60

4.4 • Two-Digit Divisors

OUT OF 22

Warm-Up • 1-12 • 2 | 2 | 3 | 4 | 3 | 3 | 2 | 4 | 4 | 4 | 9 | 7

Core • 13-22 • 7 | 6 | 16 | 15 | 23 | 76 | 76 | 76 | 76 | 76

Challenge • 23-26 • 23 | 27 | 76 | 76

Fri • Perfect Numbers

OUT OF 8

Warm-Up • 1-4 • 6 | 7 | 3 | 5

Core • 5-8 • 28 | 16 | 63 | 128

Challenge • 9-14 • 28 | 496 | 496 | 168 | 144 | 55

Missing-Digit Division

Reason backwards from the quotient to the digits under the bar.

Warm-Up 12 ITEMS • RECALL

1 $? \div 4 = 6$

2 $? \div 7 = 8$

3 $36 \div ? = 9$

4 $81 \div ? = 9$

5 $? \div 5 = 12$

6 $48 \div ? = 6$

7 $8 \div 4$

8 $18 \div 3$

9 $35 \div 7$

10 $48 \div 12$

11 $55 \div 5$

12 $72 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $? \div 6 = 103$

14. $372 \div ? = 124$

15. $? \div 8 = 91$

16. $? \div 12 = 15$

17. $595 \div ? = 119$

18. $126 \div 6$

19. $203 \div 7$

20. $288 \div 8$

21. $351 \div 9$

22. $425 \div 5$



Working space



Error journal

Snack Run

Real quantities, real people, every leftover call written down.

Warm-Up 12 ITEMS • RECALL

1 30 crackers between 5 — each

2 24 grapes between 4 — each

3 18 between 6 — each

4 25 between 5 — each

5 40 between 8 — each

6 21 between 7 — each

7 $25 \div 5$

8 $42 \div 6$

9 $56 \div 8$

10 $63 \div 7$

11 $81 \div 9$

12 $100 \div 4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. A 44-pack between 6 — whole each

.....

14. A 44-pack between 6 — leftover

.....

15. \$18 split 4 ways — dollars each

.....

16. You need 5 each for 9 people. Packs of 12 needed

.....

17. You need 5 each for 9 people. Packs of 12 needed — snacks left over

.....

18. $175 \div 5$

.....

19. $216 \div 4$

.....

20. $301 \div 7$

.....

21. $464 \div 8$

.....

22. $531 \div 9$

.....

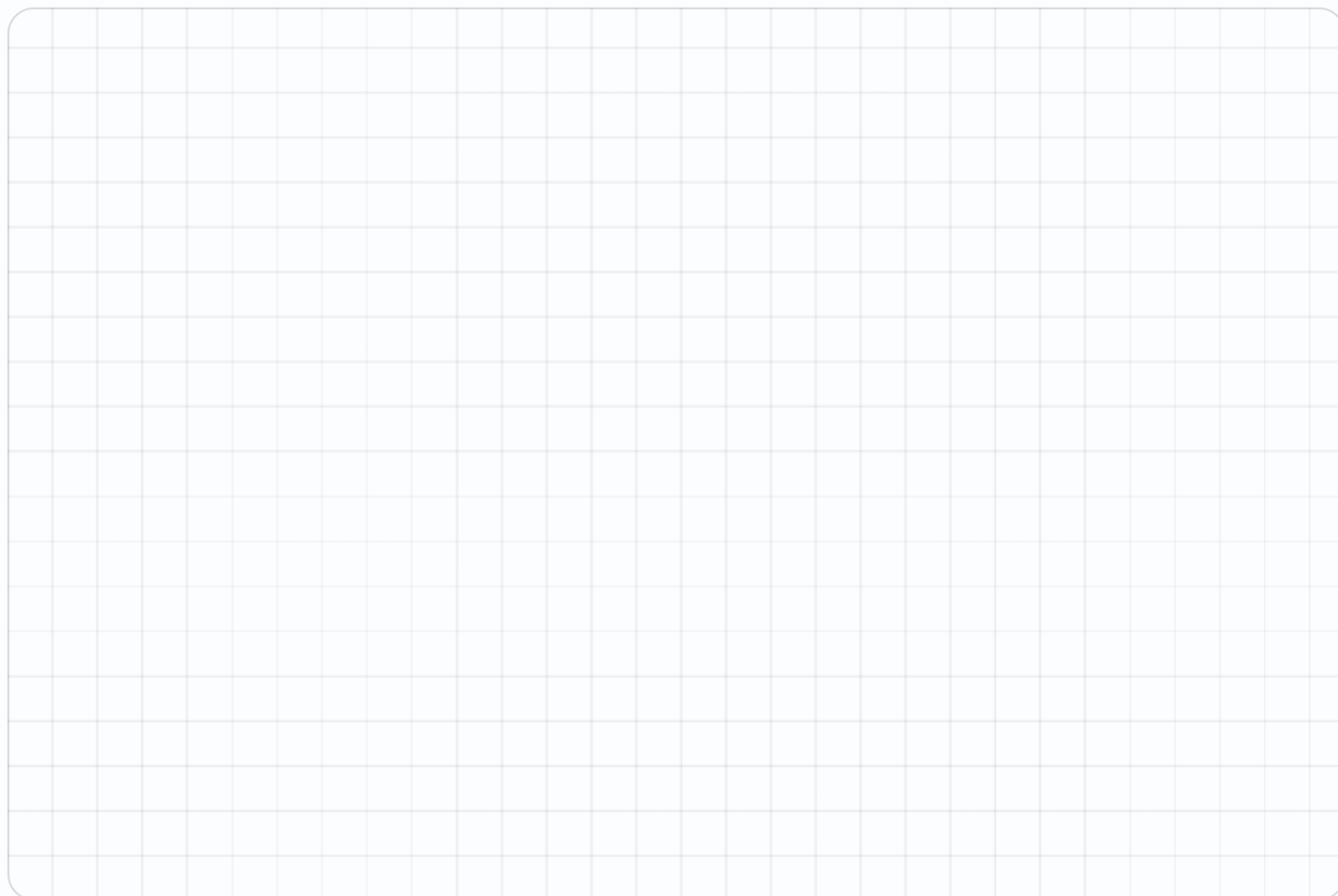
★ Challenge NOT GRADED • REASON IT OUT

23. \$25 buys packs at \$4 — whole packs

24. That purchase — dollars change

25. $2400 \div 16$

26. $3150 \div 45$

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Remainder Defence

One division, four right answers. Say which one the question wanted.

Warm-Up 10 ITEMS • RECALL

1 $27 \div 4$ — the whole part

2 27 children in cars of 4 — cars needed

3 27 cookies between 4 — whole each

4 $27 \div 4$ rounded down

5 $30 \div 5$

6 $44 \div 4$

7 $66 \div 6$

8 $77 \div 7$

9 $88 \div 8$

10 $99 \div 9$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $53 \div 8$ — the whole part

.....

12. 53 people in vans of 8 — vans needed

.....

13. \$53 shared by 8 — whole dollars each

.....

14. $132 \div 4$

.....

15. $264 \div 8$

.....

16. $396 \div 12$

.....

17. $231 \div 7$

.....

18. $165 \div 5$

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

Warm-Up 10 ITEMS • RECALL

1 $63 \div 9$

2 $120 \div 4$

3 Is 45 divisible by 9 — yes or no

4 $81 \div 9$

5 $200 \div 5$

6 $10 \div 2$

7 $16 \div 2$

8 $21 \div 3$

9 $32 \div 8$

10 $50 \div 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $456 \div 4$

.....

12. $618 \div 6$

.....

13. $3,472 \div 8$

.....

14. $84 \div 12$

.....

15. $294 \div 6$

.....

16. $343 \div 7$

.....

17. $392 \div 8$

.....

18. $441 \div 9$

.....

19. $245 \div 5$

.....

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 $48 \div 6$

2 $150 \div 5$

3 $24 \div 4$

4 $40 \div 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $372 \div 3$

6. $728 \div 8$

7. $915 \div 3$

8. $3,472 \div 8$

9. $96 \div 16$

10. Is 5,346 divisible by 9 — yes or no

11. 127 children, buses of 30 — buses needed

12. $156 \div 6$

13. $322 \div 7$

 Challenge NOT GRADED • REASON IT OUT

14. $26 \div 4$ as a decimal

15. $945 \div 35$

16. A quotient of 23 remainder 5 with divisor 9 — the dividend

17. $1728 \div 24$

18. $4356 \div 66$

19. $5000 \div 40$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Missing-Digit Division

OUT OF 22

Warm-Up • 1-12 • 24 | 56 | 4 | 9 | 60 | 8 | 2 | 6 | 5 | 4 | 11 | 8

Core • 13-22 • 618 | 3 | 728 | 180 | 5 | 21 | 29 | 36 | 39 | 85

Challenge • 23-26 • 301 | 426 | 153 | 63

5.2 • Snack Run

OUT OF 22

Warm-Up • 1-12 • 6 | 6 | 3 | 5 | 5 | 3 | 5 | 7 | 7 | 9 | 9 | 25

Core • 13-22 • 7 | 2 | 4.5 | 4 | 3 | 35 | 54 | 43 | 58 | 59

Challenge • 23-26 • 6 | 1 | 150 | 70

5.3 • Remainder Defence

OUT OF 18

Warm-Up • 1-10 • 6 | 7 | 6 | 6 | 6 | 11 | 11 | 11 | 11 | 11

Core • 11-18 • 6 | 7 | 6 | 33 | 33 | 33 | 33 | 33

Challenge • 19-26 • 3 | 6.75 | 6.625 | 5 | 6 | 10 | 33 | 33

Thu • Error Journal Sweep

OUT OF 21

Warm-Up • 1-11 • 7 | 30 | yes | 2 | 9 | 40 | 5 | 8 | 7 | 4 | 10

Core • 12-21 • 114 | 103 | 434 | 7 | 2 | 49 | 49 | 49 | 49 | 49

Challenge • 22-25 • 23 | 426 | 49 | 49

Fri • Mission 02 Test

OUT OF 14

Warm-Up • 1-4 • 8 | 30 | 6 | 8

Core • 5-14 • 124 | 91 | 305 | 434 | 6 | yes | 5 | 6.5 | 26 | 46

Challenge • 15-19 • 27 | 212 | 72 | 66 | 125